

TAMILS AND TECHNOLOGY MODEL QUESTION PAPER- ANSWERS

PART-A

1. Which raw material is used in Philippines for making clothes? **Ans: Banana fiber.**
2. How did Herodotus describe Indian cotton? **Ans: Cotton growing on trees.**
3. Apart from cotton, from which trees fiber collected for weaving **Ans: Ilavamaram and Kongumaram.**
4. In what were the threads wound on the spinning wheel first stored? **Ans: Kathir**
5. Total number of arts in which Tamils were excelled is. **Ans: 64**
6. Buildings were classified as Chittril(Small house) and Peril(Big house) based on. **Ans: The architectural design of the house.**
7. The size of the design for constructing the dance stage is. **Ans:24 hand finger prints.**
8. Which is the largest among the monolithic chariots? **Ans: Draupathi ratha**
9. Before the time of Tolkapiyam, the tradition of obtaining resources is based on. **Ans: Sea based resource gathering tradition.**
10. Which metal played an important role in advancing human civilization from the Stone age to the Metal age? **Ans: Iron**
11. The main device used to steer a ship in right direction is **Ans: Sukkan**
12. Poozhiyar refers to . **Ans: Collectors of raw materials for beads.**
13. What is the fundamental basis for Agricultural industry? **Ans: Water**
14. Waterfalls and springs refers to which water body type? **Ans: Kurinji**
15. What is the name of the outlet used to release the surplus water? **Ans: Kalingu**
16. The plough used by ancient tamils was shaped like. **Ans: Mouth of a female elephant.**
17. The poet who stated Galileo's scientific truth that a distant object can be made to appear as if it is near. **Ans: Kambar**
18. Who served as the founding Vice-Chancellor of the Tamil virtual academy? **Ans: Kulandaisamy**
19. Statement 1: A computer is an electronic device.
Statement 2: It operates based on machine language. **Ans: Both Statement 1 and 2 are correct.**
20. How many books are there in Roja Muthiah research library? **Ans: Over 1,00,000.**

PART-B

21. What are the types of rotators?

Ball-headed Rotator (Kundu Munai), Conical Pit Rotator (Koombu Kuzhi)

22. Write any two cave temples in Mamallapuram.

Varaha Cave Temple, Mahishasuramardini Cave Temple

23. What are the metals used for making coins in the Sangam age?

Copper, Silver, Iron and Gold

24. Write about Natural Fertilizer.

- Derived from organic sources like **Green Manure** (leaves), **Animal Waste** (cow dung), and **Compost**.
- Enriches soil nutrients naturally without causing chemical damage to the environment.

25. Define bilingual code.

- A system that allows computers to process **two languages** (like Tamil and English) simultaneously.
- Uses a single character set or framework, such as **Unicode** or **TSCII**.

26. What is sales tax?

- An **indirect tax** levied by the government on the sale of goods and services.
- It is collected by the retailer from the consumer and then remitted to the government.

27. Write about Chettinad Architecture.

- Features large **open courtyards** and heavy pillars made of **Burma teak**.
- Uses **Athangudi tiles** for flooring and a unique **lime plaster** (eggshell finish) for cool, shiny walls.

28. What is the function of encoding?

- Converts human-readable data into a **digital format** (like Binary) that computers can process.
- Ensures information can be **stored and transmitted** efficiently across different systems

PART-C

29. Describe the process of pottery making.

1. Potters were called '**Kuyavar**'
2. The tradition of making pottery is referred to as '**Panai Vanaithal**'
3. Pottery making is the oldest and foremost handicraft technology of the Tamils.
4. High-quality clay required for pottery is collected from riverbeds or ponds.
5. The collected clay is mixed with sand and water, and then well-processed by treading it with feet.
6. The processed clay is placed in the center of a rotating wheel called 'Thiruvai'.
7. As the wheel spins, gentle pressure is applied with the hands to create the desired shapes (pots, pans).
8. The bottom of the pot removed from the wheel remains open; it is closed and leveled by tapping it with a wooden beater (Thatti).
9. The designed pots are first dried in the shade and then thoroughly dried in the sun.
10. To gain strength, the dried pots are stacked in a firing kiln called 'Soolai'.
11. Due to firing at high temperatures, the pottery becomes strong and attains red or black colors.
12. The 'Black-and-Red' pottery technology used by the Tamils is a testament to their unique knowledge.

30. Describe the model structure of ThirumalaiNayakarMahal.

1. Thirumalai Nayakkar Mahal was built in Madurai in 1636 AD by King Thirumalai Nayak.
2. It is built in the 'Indo-Saracenic' architectural style, which is a blend of Dravidian and Islamic arts.
3. The most outstanding feature of this palace is its magnificent pillars.
4. The massive pillars, approximately 82 feet high and 19 feet in circumference, give majesty to the palace.

5. The pillars and walls are coated with a plaster made of lime and egg white, which still shines like marble today.
6. The Durbar Hall, known as 'SwargaVilasam', is the most important part of the palace.
7. Its central dome is an engineering marvel, built using only bricks without any iron or wooden supports.
8. Intricate Yali sculptures and artistic paintings are carved on the arches and roof sections.
9. The open courtyard of the palace served as a venue for cultural performances and festivals in those days.
10. Preserved today by the Tamil Nadu Archaeological Department, this palace proclaims history through the daily 'Sound and Light' show.

31. Explain about gemstone types as described in Silappathigaram.

1. While singing about the 'Street of Gem Merchants', IlangoAdigal mentions nine types of gems.
2. Diamonds must be free from four flaws: crow's foot (Kakkapatham), dirt (Kalangam), spot (Vindhu), and lines (Iregai).
3. In emeralds, the absence of water-lines (Neerottam) and darkness that hides light indicates high quality.
4. Rubies are of four types: Pathumam, Neelam, Vintham, and Padigam; they are special for emitting five types of light.
5. Topaz (Pushparagam) should have brilliance like sunlight and a golden light like honey.
6. Cat's eye (Vaiduriyam) has a light like a cat's eye and a glow mixed with golden minerals.
7. Sapphire (Neelamani) is unique for emitting the dark light of rain clouds.
8. Garnet (Komedhagam) possesses a distinct light mixed with honey and red colors.
9. Corals must be in a bright red color, free from natural defects like splits, bends, and twists.
10. The book confirms that Tamils excelled globally in removing flaws and grading the quality of these nine gems.

32. Describe the medicinal property of pearl.

1. **MuthuSalabam:** Diving into the sea to harvest pearls.
2. **Salabam:** The term for the place in the sea where pearls constitute/grow.
3. **PavalaSalabam:** Diving into the sea to harvest coral
4. Pearls have a cooling nature; they reduce increased 'Pitta' in the body and balance body heat.
5. They relieve stress, provide peace of mind, and serve as an excellent medicine for insomnia.
6. Since they are rich in natural calcium, they greatly help in the strength of bones and teeth.
7. Pearl Bhasma is used to relieve eye irritation and to provide glow to the skin to preserve youthfulness.
8. A pearl formed around a sand grain or parasite entering an oyster is a remarkable natural defense technology.
9. Pure white and lustrous pearls are praised in Silappathigaram as high-quality gems without any flaws.

10. Pearl Churnam and Bhasma help in curing anemia and regulating the nervous system.
11. For thousands of years, Tamils were world pioneers in pearl diving and its medicinal applications.
12. Beyond being an ornament, the pearl is still revered today as a superior natural medicine that enhances physical and mental health.

33. Write about the objectives and advantages of Sorkuvaischeme.

1. This scheme was launched by the Government of Tamil Nadu to increase the vocabulary of the Tamil language and protect pure Tamil words.
2. Its primary objective is to retrieve and bring back into use ancient Tamil words that are disappearing.
3. It creates new Tamil technical terms equivalent to those in growing science and modern technology.
4. The words created through this scheme help in understanding complex scientific concepts easily in Tamil itself.
5. It prevents the unnecessary mixture of foreign words and protects the uniqueness and purity of the Tamil language.
6. It serves as a massive 'Digital Dictionary' on the website for students, teachers, and researchers.
7. This scheme serves as an excellent platform for the public and Tamil scholars to suggest new words they create.
8. It categorizes technical terms in every field (Medicine, Engineering, Computer, etc.) and provides them alphabetically.
9. This scheme unites Tamils living worldwide linguistically and takes the development of the Tamil language to global standards.
10. This scheme acts as a backbone for the Tamil language to adapt itself to changing times and remain a vibrant language.